

SQL Query Execution Order

Easy Reference & Exam Guide – GATE Smashers Summary

Fast Exam Trick (Mnemonic)

Memorize the sequence of initial letters for logical execution order

F J W G H S D O L

Key Concept: SQL queries are *written* in one order (e.g., starting with `SELECT`), but RDBMS engines *logically execute* them in a completely different order (starting with `FROM`).

Detailed Logical Execution Steps (1 to 5)

1

FROM

Identifies the source table(s) from which the primary dataset is retrieved.

`FROM employee`

2

JOIN

Combines rows from secondary tables based on specified join conditions.

`JOIN department ON employee.dept_id = department.dept_id`

3

WHERE

Filters individual **rows** based on predicate conditions before grouping occurs.

`WHERE city = 'Delhi'`

4

GROUP BY

Aggregates rows sharing identical values in specified columns into distinct summary groups.

`GROUP BY department`

5

HAVING

Filters entire **groups** created by `GROUP BY` using aggregate functions.

`HAVING COUNT(department) < 5`

Crucial Distinction:

- `WHERE` filters individual **rows** before aggregation takes place.
- `HAVING` filters summarized **groups** after aggregation takes place.

Detailed Logical Execution Steps (6 to 9)

6	SELECT Computes expressions and determines which specific columns appear in the result set.	SELECT department, COUNT(department)
7	DISTINCT Evaluates the selected columns and eliminates duplicate rows from the output dataset.	SELECT DISTINCT department
8	ORDER BY Sorts the final result set in ascending (ASC) or descending (DESC) order.	ORDER BY COUNT(department) DESC
9	LIMIT Restricts the maximum number of rows returned by the query engine.	LIMIT 1

Comprehensive Query Example

```
-- Full SQL Query Example
SELECT department, COUNT(department) FROM employee WHERE
```

Execution Order Comparison

Phase	Written Order (Syntax)	Logical Order (Execution)
1st	SELECT	FROM / JOIN
2nd	FROM / JOIN	WHERE
3rd	WHERE	GROUP BY
4th	GROUP BY	HAVING
5th	HAVING	SELECT
6th	ORDER BY	DISTINCT
7th	LIMIT	ORDER BY → LIMIT